User Need and AI Contribution

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected need and capstone-planning record.

Explain the service, trace the corpus boundary, audit claims and plan a revealing test.

Keep one habit: Keep user task, baseline, proposed contribution and unmeasured outcomes explicit.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Retrieval Augmented Generation

Verify Numbers and Dates

Multimodal Evidence

W1 · Explain a bounded capstone need

> Name user, task, evidence and fallback.

Your evidence record

- Room-assistant planning practice: public register P, teacher-only guide T and current approval A; student service reports approved room capacity or needs_review. Define eligible corpus P+A, manual same-room approval/capacity lookup baseline and complete/missing/conflicting test cases. No actual room capacity or performance result is supplied.

  Separate glossary need: G3-v1 contains only habitat, meaning the place an organism lives; photosynthesis absent; compare direct lookup with a bounded restatement helper.

- Reference for this exercise: open this week's student lesson, "User Need and AI Contribution", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the room-assistant plan

> Keep permissions, baseline and reserved cases visible.

Your evidence record

- Room-assistant planning practice: public register P, teacher-only guide T and current approval A; student service reports approved room capacity or needs_review. Define eligible corpus P+A, manual same-room approval/capacity lookup baseline and complete/missing/conflicting test cases. No actual room capacity or performance result is supplied.

  Separate glossary need: G3-v1 contains only habitat, meaning the place an organism lives; photosynthesis absent; compare direct lookup with a bounded restatement helper.

- Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or **need**s_review. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Audit a plan-as-proof claim

> Requirements do not demonstrate implementation success or learner benefit.

Your evidence record

- Room-assistant planning practice: public register P, teacher-only guide T and current approval A; student service reports approved room capacity or needs_review. Define eligible corpus P+A, manual same-room approval/capacity lookup baseline and complete/missing/conflicting test cases. No actual room capacity or performance result is supplied.

  Separate glossary need: G3-v1 contains only habitat, meaning the place an organism lives; photosynthesis absent; compare direct lookup with a bounded restatement helper.

- A report omits this qualification: A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing need check

> Specify a complete test and expected evidence before building.

Your evidence record

- Room-assistant planning practice: public register P, teacher-only guide T and current approval A; student service reports approved room capacity or needs_review. Define eligible corpus P+A, manual same-room approval/capacity lookup baseline and complete/missing/conflicting test cases. No actual room capacity or performance result is supplied.

  Separate glossary need: G3-v1 contains only habitat, meaning the place an organism lives; photosynthesis absent; compare direct lookup with a bounded restatement helper.

- Reference for this exercise: open this week's student lesson, "User Need and AI Contribution", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would **need** to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Scope the equipment assistant independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Plan the equipment assistant using only the stated public/restricted labels and question. No inventory quantities, booking decisions or live trial results are supplied; this assessment asks for a corpus boundary, baseline and missing-evidence test rather than a fabricated booking answer.

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-**evidence** test.

> **My independent explanation, trace and limits**
>
> _____
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> _____
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> _____
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> _____

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Justify exactly which Study Buddy stage uses AI and which stages remain deterministic.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Capstone **requirement**s and corpus. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Capstone **requirement**s and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if **need**ed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Students cannot find definitions in a 20-page handout. Compare an index with a grounded assistant. Define a success measure and a failure case for both.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Learners struggle to locate definitions in an approved chapter; a proposed chatbot answers everything.

Narrow the **need** and **AI contribution**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A student **need**s the exact due date from a course notice. Should the design start by generating a plausible date or locating the notice? Explain the **AI contribution**.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 19 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Students cannot find definitions in a 20-page handout. Compare an index with a grounded assistant. Define a success measure and a failure case for both.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-**evidence** test.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

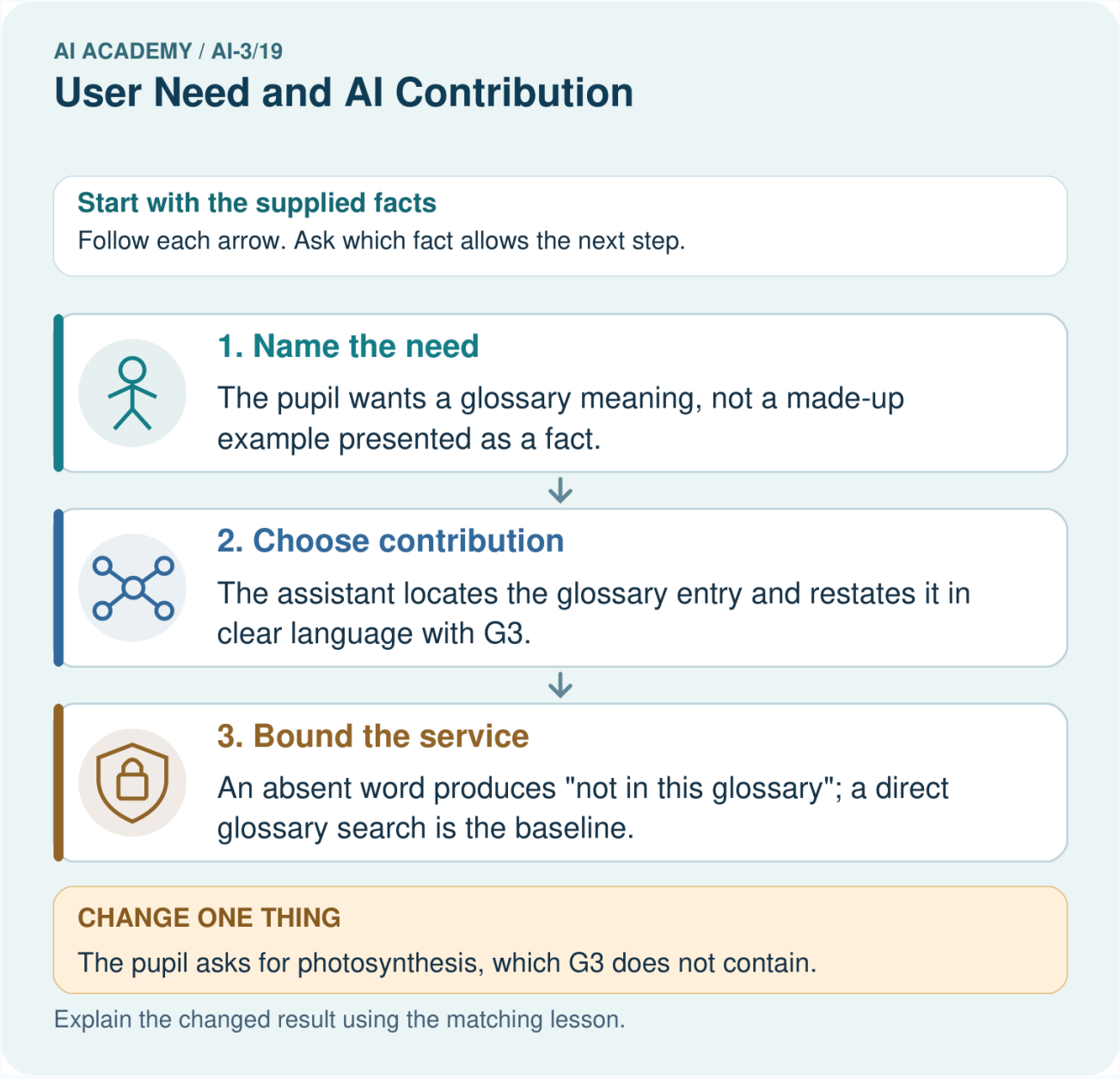
> Keep the explanation and the support you used.

Use this when helpful

Define a useful reading helper before choosing its AI features.

A pupil **need**s the meaning of unfamiliar words in an approved glossary. G3 defines habitat as the place an organism lives; no other sources are allowed.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Define a useful reading helper before choosing its AI features.
- Which labelled step matches this detail? An absent word produces "not in this glossary"; a direct glossary search is the baseline.
- 1. Bound the service

- 2. Name the **need**
- 3. Choose contribution
- Hint if **need**ed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Bound the service An absent word produces “not in this glossary”; a direct glossary search is the baseline. Read the picture in order.

  An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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> ---
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> ---
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> ---

▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: The pupil asks for photosynthesis, which G3 does not contain. Which conclusion is supported by the supplied facts?
- 1. The assistant reports that the definition is unavailable in its approved glossary.

- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if **need**ed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: The assistant reports that the definition is unavailable in its approved glossary. A helpful interface cannot expand its authorised **evidence** by guessing.

> **My selected conclusion, deciding fact and explanation**
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> ---
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▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the **mechanism** and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Verify Numbers and Dates (ai-3/16). Goal: Verify a numeric claim using its source, denominator and units. Short practice: Which part can be corrected immediately from existing **evidence**, and which **need**s follow-up?. Return to this week and explain what you changed.

Review Multimodal **Evidence** (ai-3/13). Goal: Keep visible **evidence**, hidden contents and caption claims separate. Short practice: Why were both “exactly three” and “exactly five” unsupported before opening the box?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

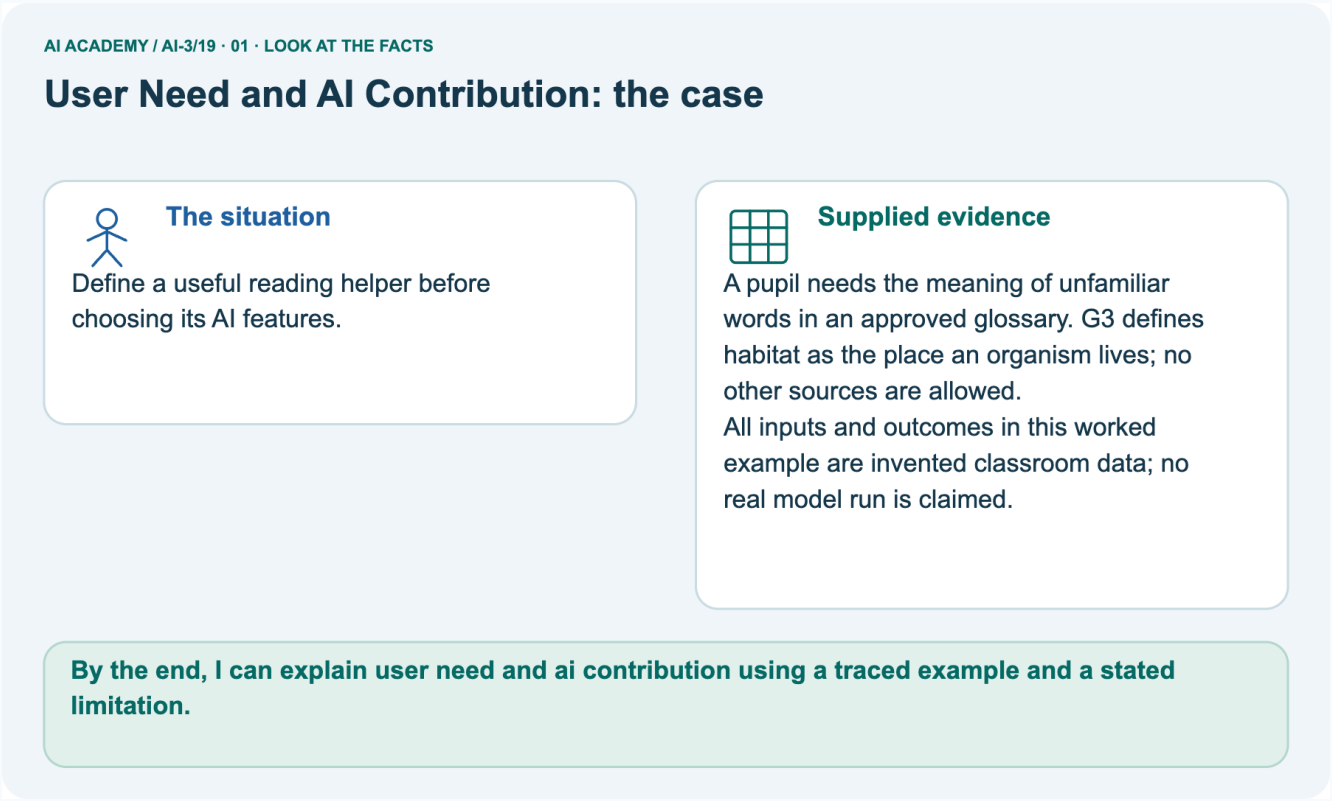
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. User Need and AI Contribution: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

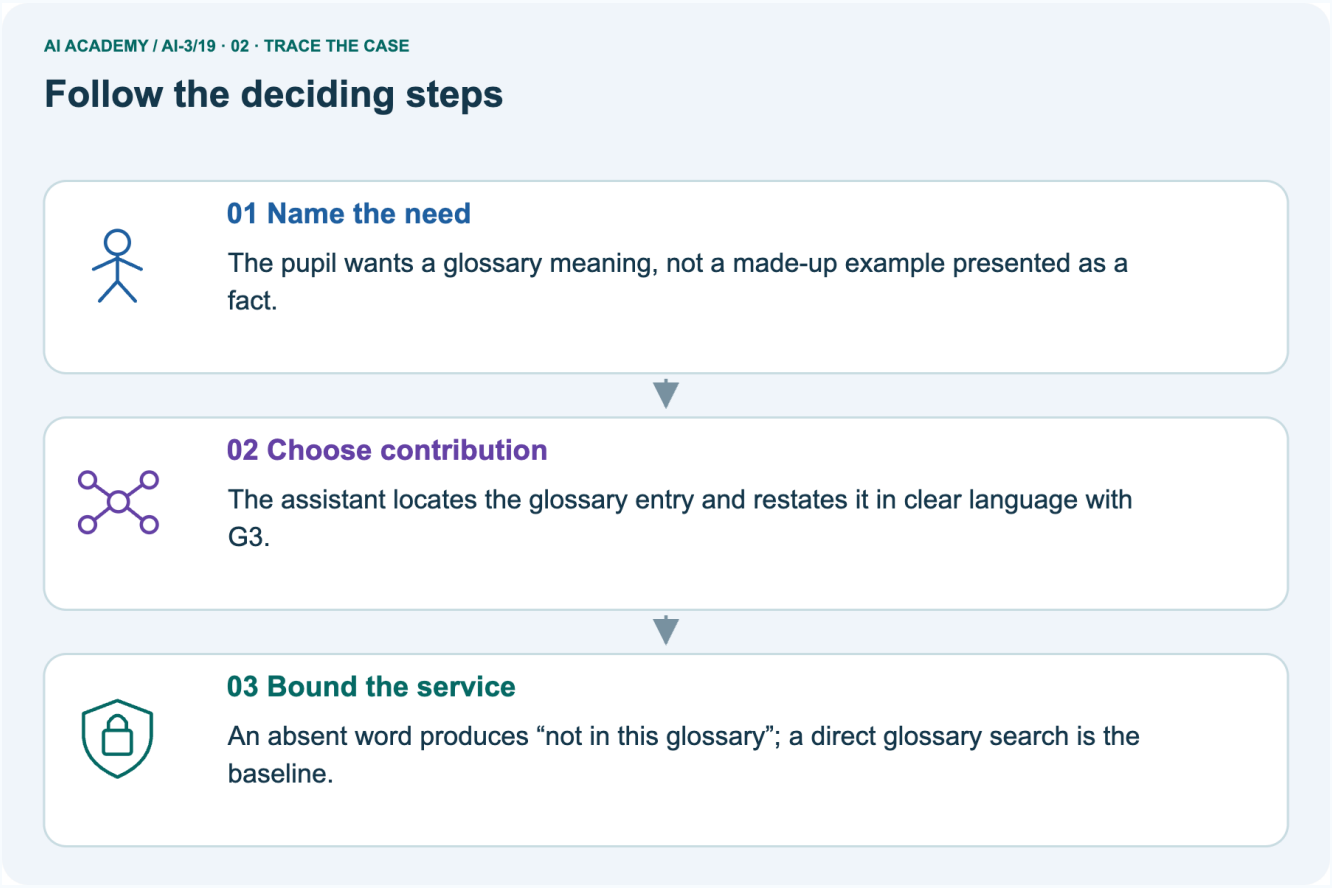
Define a useful reading helper before choosing its AI features. Supplied facts: A pupil **need**s the meaning of unfamiliar words in an approved glossary.

G3 defines habitat as the place an organism lives; no other sources are allowed. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



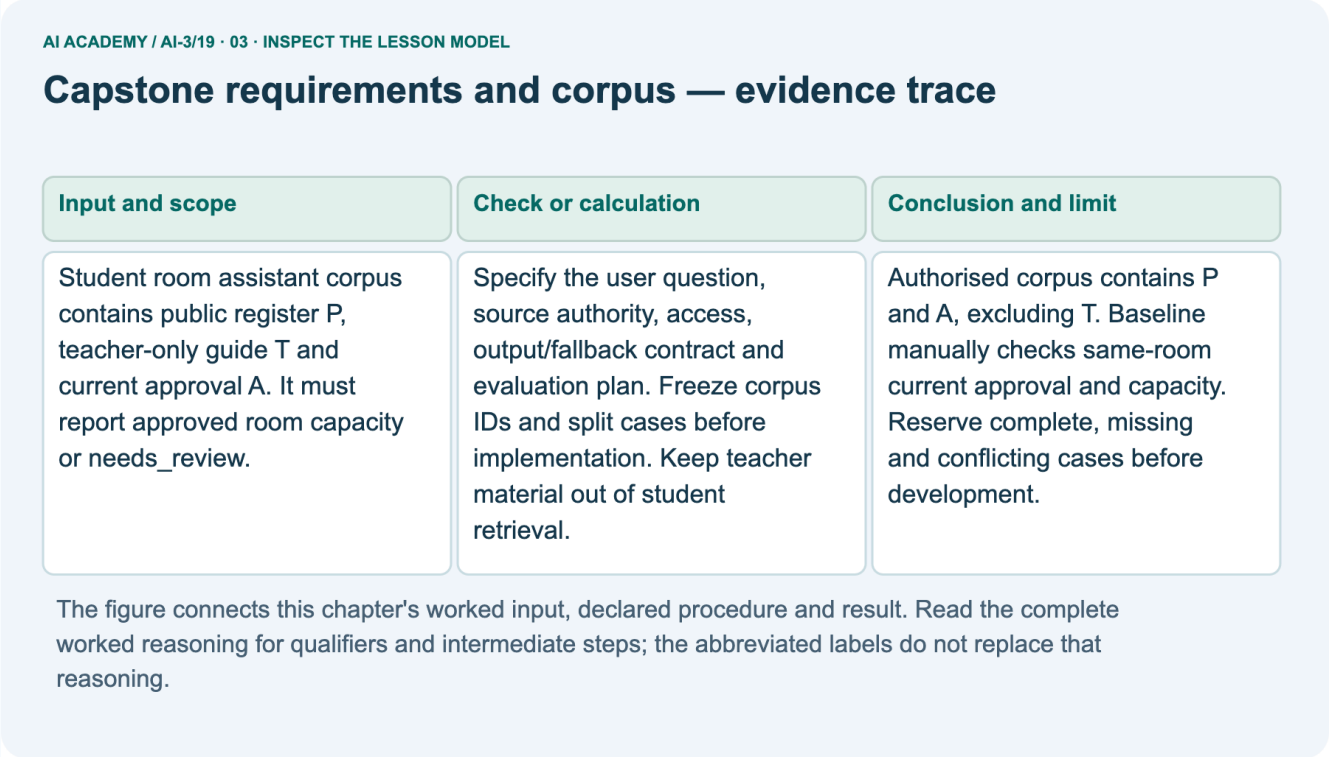
Follow the labelled picture with the matching explanation. Open the vector diagram

Name the **need**: The pupil wants a glossary meaning, not a made-up example presented as a fact. → Choose contribution: The assistant locates the glossary entry and restates it in clear language with G3. → Bound the service: An absent word produces “not in this glossary”; a direct glossary search is the baseline.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone requirements and corpus — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **requirement**s and corpus — **evidence** trace. Student room assistant corpus contains public register P, teacher-only guide T and current approval A.

It must report approved room capacity or **need**s_review. | Specify the **user** question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation.

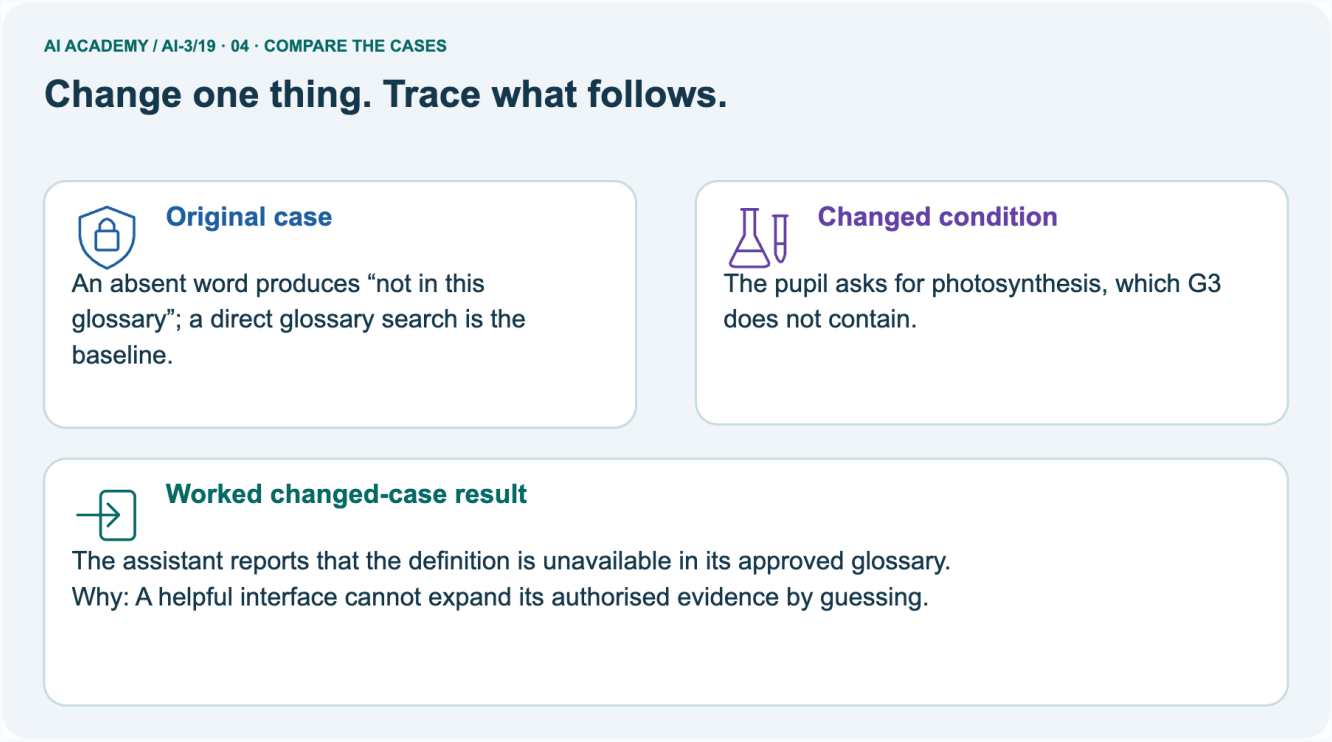
Keep teacher material out of student retrieval. | Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity.

Reserve complete, missing and conflicting cases before development.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

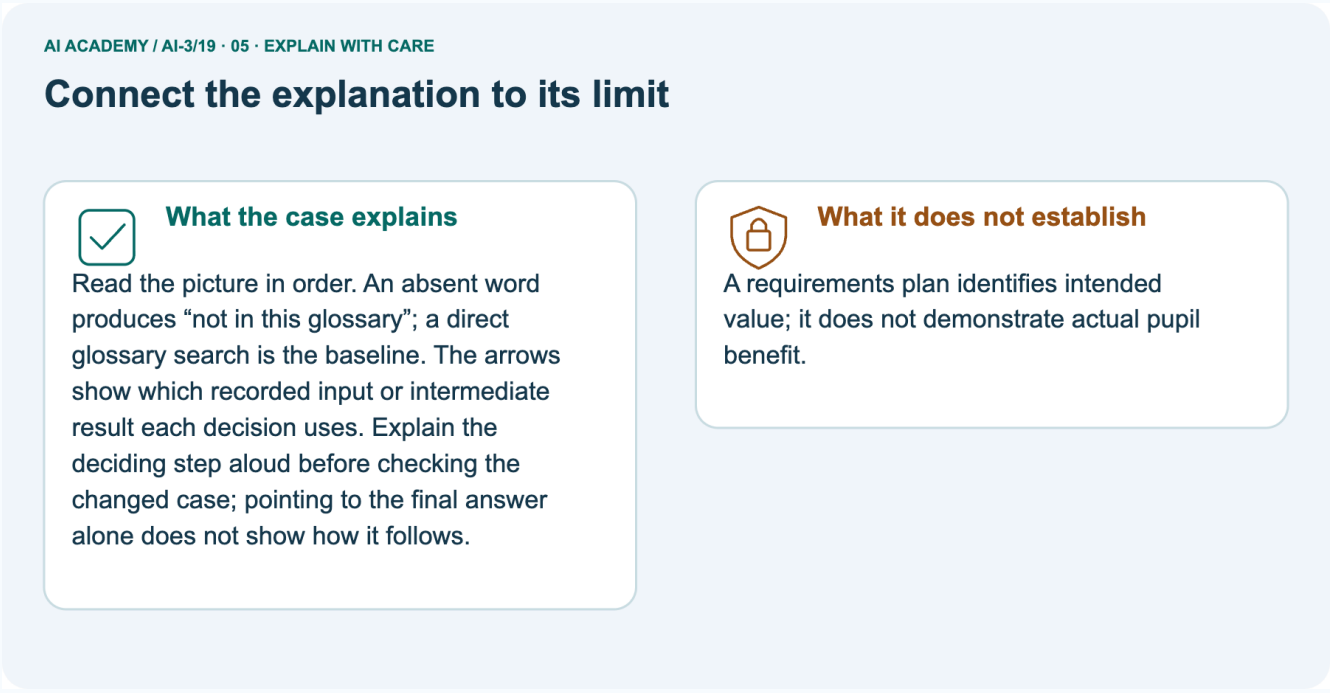
Original case: An absent word produces “not in this glossary”; a direct glossary search is the baseline. Changed condition: The pupil asks for photosynthesis, which G3 does not contain.

Expected result: The assistant reports that the definition is unavailable in its approved glossary. Reason: A helpful interface cannot expand its authorised **evidence** by guessing.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded **input** or intermediate result each decision uses.

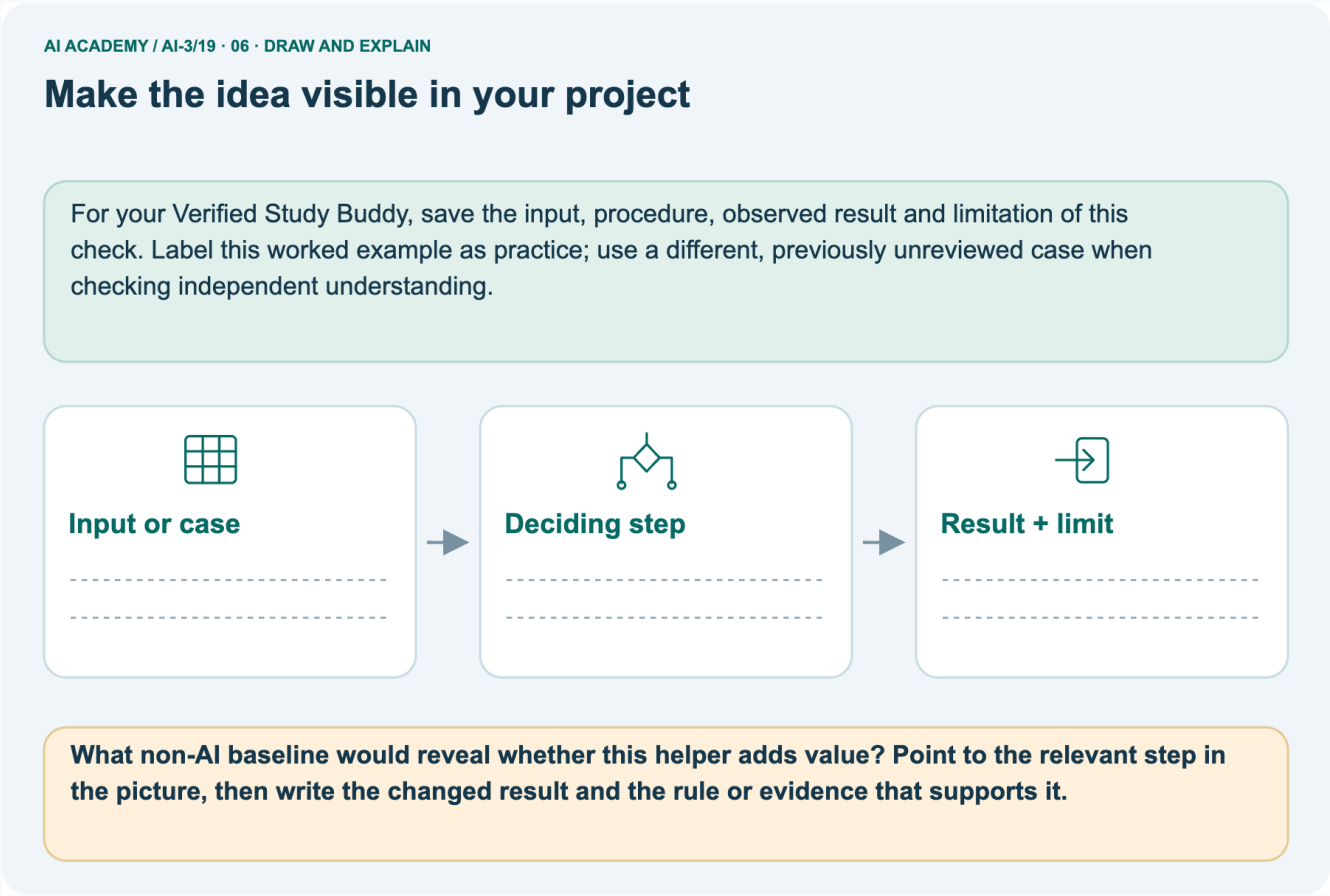
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A **requirement**s plan identifies intended value; it does not demonstrate actual pupil benefit.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**
>
> _____
>
> _____
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> _____
>
> _____

▼ **Earlier wording, alternatives and picture transcripts**

01. User Need and AI Contribution: the case

02. Follow the deciding steps

03. Capstone requirements and corpus — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.